KERALA POLYTECHNIC · REVISION 2026

Digital Computer Fundamentals

Complete student lesson for course code **3134**.

Revision 2026 Semester 3 Program Core 4 modules

Course snapshot

Scheme: 3 (L:3,T:0,P:0); 45 periods

Credits: 3

Contact: 3 (L:3,T:0,P:0)

Module hours listed: 43

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

3134

Semester

3

Credits

3

Teaching scheme

3 (L:3,T:0,P:0); 45 periods

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Number systems and digital codes	8	CO1 · Applying
2	Boolean algebra and Karnaugh maps	11	CO2 · Applying
3	Combinational circuits	12	CO3 · Applying
4	Sequential circuits	12	CO4 · Applying

Prerequisites

- Fundamentals of Electrical and Electronics; semiconductor physics, diodes, transistors and rectifiers.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO–PO Information

Official course objectives

1. Understand the data representation in the Computer System.
2. Perform number conversion from one system to another.
3. Understand use of boolean algebra and K-Map in digital circuits.
4. Equip students to design and develop simple combinational and sequential circuits.

Course Outcomes

1. Perform number system conversions, binary arithmetic operations and binary coding.
2. Make use of Boolean algebra and the Karnaugh Map for the implementation of logic functions.
3. Design combinational circuits.
4. Design sequential circuits.

CO–PO mapping as supplied

CO–PO mapping is reproduced from the supplied syllabus header; 3 = strongly mapped, 2 = moderately mapped, 1 = weakly mapped, and a blank cell is not silently converted into a value.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Number systems and base conversion	3
module-1	M1.02	Complements and binary arithmetic	3
module-1	M1.03	Binary codes and data representation	2
module-2	M2.01	Boolean laws and logic gates	4
module-2	M2.02	Canonical forms, minterms and maxterms	3
module-2	M2.03	Karnaugh-map simplification	4
module-3	M3.01	Half and full adders	3
module-3	M3.02	Subtractors and arithmetic circuits	3
module-3	M3.03	Encoders, decoders and comparators	3
module-3	M3.04	Multiplexers and demultiplexers	3
module-4	M4.01	Latches and flip-flops	3
module-4	M4.02	Registers and shift registers	3
module-4	M4.03	Counters and frequency division	3
module-4	M4.04	Timing and sequential-system design	3

Learning Roadmap

1. Number systems and digital codes
CO1 · Applying · 8 hours

2. Boolean algebra and Karnaugh maps
CO2 · Applying · 11 hours

3. Combinational circuits
CO3 · Applying · 12 hours

4. Sequential circuits
CO4 · Applying · 12 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Number systems and digital codes

Data representation is the foundation for digital hardware. The official content covers decimal, binary, octal and hexadecimal systems; conversions; signed and unsigned representations; complements; binary arithmetic; BCD, Gray and ASCII codes.

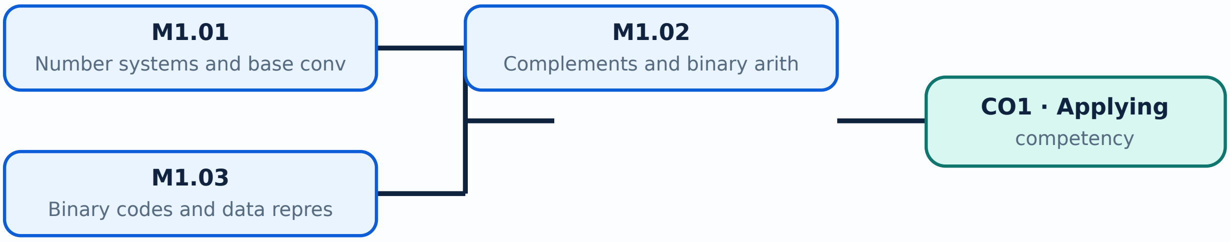
Hours: 8
CO1 · Applying · Bloom level: Applying

Learning goals

- Convert numbers between common bases.
- Perform binary arithmetic and identify suitable digital codes.
- Check a conversion by reversing the operation.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

M1.01 — Number systems and base conversion (3 hours)

Concept and explanation

A number system is defined by its radix and set of digits. Decimal, binary, octal and hexadecimal representations describe the same quantity with different symbols; repeated division or positional expansion provides systematic conversion.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- Use positional weights.
- Convert integer and fractional values carefully.
- Write the radix when ambiguity is possible.

> **Engineering / workplace application:** Microprocessors and memory addresses commonly use hexadecimal notation while logic circuits use binary states.

M1.02 — Complements and binary arithmetic (3 hours)

Concept and explanation

One’s complement and two’s complement provide signed representation and subtraction methods. Binary addition, subtraction, multiplication and division follow positional rules, with carry and borrow handled at each bit position.

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Study points

- Distinguish one’s and two’s complement.
- Check overflow in fixed-width arithmetic.
- Keep the word length consistent.

Illustrative example: For an n-bit two’s-complement number, negate it by complementing all bits and adding 1, then discard a carry beyond the word length.

M1.03 — Binary codes and data representation (2 hours)

Concept and explanation

BCD represents decimal digits separately, Gray code changes one bit between adjacent values, and ASCII represents characters. Code selection depends on the required reliability, display, communication or storage function.

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Study points

- Explain BCD, Gray and ASCII.
- Convert a small value into a code.
- State one engineering use for each code.

Engineering / workplace application: Encoders, displays, keyboards and digital communication interfaces use coded representations.

Module summary: A reliable conversion method prevents errors when a digital quantity is represented in another base.

OFFICIAL MODULE 2

Boolean algebra and Karnaugh maps

Boolean algebra models logic with variables that take 0 or 1. The syllabus includes Boolean laws, logic gates, canonical forms, minterms, maxterms and Karnaugh-map simplification.

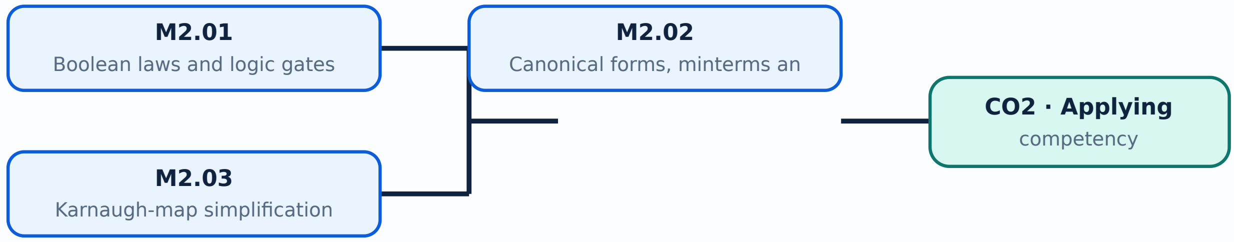
Hours: 11
CO2 · Applying · Bloom level: Applying

Learning goals

- Apply Boolean identities.
- Reduce a logic expression using a K-map.
- Translate between truth tables and expressions.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

M2.01 — Boolean laws and logic gates (4 hours)

Concept and explanation

Boolean variables and operators AND, OR and NOT form the algebra of switching circuits. De Morgan’s theorems are especially useful for changing a circuit between active-high and active-low forms.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- Use identity, null, idempotent and complement laws.
- Relate NAND and NOR to universal-gate implementation.
- Draw a truth table.

> **Engineering / workplace application:** Combinational logic in controllers and interlocks is specified using Boolean expressions.

M2.02 — Canonical forms, minterms and maxterms (3 hours)

Concept and explanation

A truth table can be expressed as a sum of minterms or product of maxterms. Canonical forms provide a standard starting point for minimization and circuit implementation.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- Identify minterms from rows containing 1.
- Identify maxterms from rows containing 0.
- Use don’t-care conditions only when justified.

> **Common mistake:** Do not interchange the row conditions for minterms and maxterms.

M2.03 — Karnaugh-map simplification (4 hours)

Concept and explanation

A Karnaugh map arranges adjacent cells in Gray-code order so groups of 1s or 0s reveal simplified terms. Groups must contain powers of two and may wrap around the map edges.

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Study points

- Make valid groups.
- Select essential prime implicants.
- Implement the reduced expression with suitable gates.

Illustrative example: For four variables, begin with the largest valid group, then cover every required 1-cell with the fewest essential groups.

Module summary: Simplification reduces gate count, delay and wiring in a practical digital circuit.

OFFICIAL MODULE 3

Combinational circuits

Combinational outputs depend only on present inputs. The official coverage includes adders, subtractors, comparators, encoders, decoders, multiplexers and demultiplexers.

Hours: 12
CO3 · Applying · Bloom level: Applying

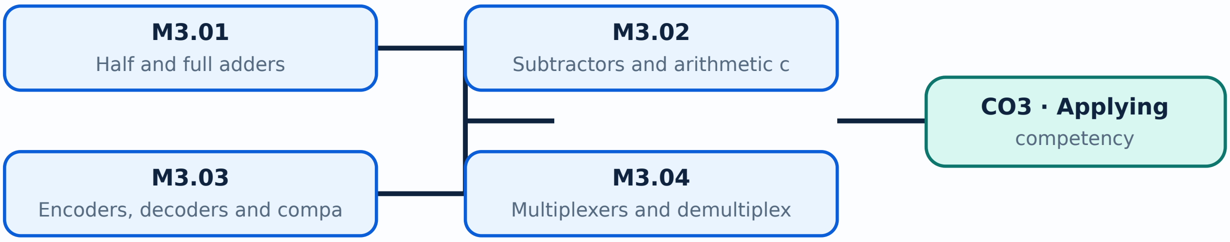
Learning goals

- Construct a truth table and logic expression.
- Choose a standard combinational block.

- Verify the output for representative input combinations.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

M3.01 — Half and full adders (3 hours)

Concept and explanation

A half adder adds two bits and produces sum and carry. A full adder adds two bits and an input carry, allowing multi-bit addition by cascading stages.

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Study points

- Write sum and carry equations.
- Explain carry propagation.
- Distinguish half and full adders.

Engineering / workplace application: Arithmetic logic units use cascaded full adders.

M3.02 — Subtractors and arithmetic circuits (3 hours)

Concept and explanation

Half and full subtractors produce difference and borrow. Complement-based subtraction can also use adder hardware, which is why arithmetic circuits often combine addition and subtraction control.

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Study points

- State difference and borrow outputs.
- Explain two’s-complement subtraction.
- Check sign and overflow conditions.

M3.03 — Encoders, decoders and comparators (3 hours)

Concept and explanation

An encoder converts one active input into a coded output, while a decoder activates one output for a coded input. A magnitude comparator indicates A greater than, equal to, or less than B.

Malayalam support: □ □□□□ □□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- Identify input/output roles.
- Use enable inputs correctly.
- Interpret priority encoders where more than one input may be active.

Engineering / workplace application: Address decoding and keypad interfaces use decoders and encoders.

M3.04 — Multiplexers and demultiplexers (3 hours)

Concept and explanation

A multiplexer selects one of many data inputs using select lines. A demultiplexer routes one input to a selected output, making both useful for data routing and function generation.

Malayalam support: □ □□□□ □□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□ □□□□□□□□□.

Study points

- Calculate the number of select lines.
- Trace the selected data path.
- Use a multiplexer to realise a Boolean function.

Module summary: A clean design proceeds from specification to truth table, simplification, circuit drawing and verification.

OFFICIAL MODULE 4

Sequential circuits

Sequential circuits include memory, so output depends on present inputs and previous state. The syllabus covers latches, flip-flops, registers and counters.

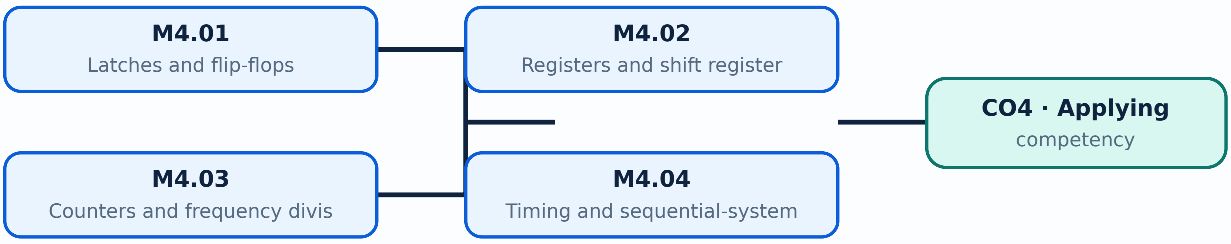
Hours: 12
CO4 · Applying · Bloom level: Applying

Learning goals

- Explain the role of a clock and state.
- Read characteristic and excitation tables.
- Select a register or counter arrangement.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

M4.01 — Latches and flip-flops (3 hours)

Concept and explanation

A latch is level-sensitive whereas a flip-flop is normally edge-triggered. SR, JK, D and T forms provide different ways to store and change a single bit.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□ □□□□□□□□□.

Study points

- Compare level and edge triggering.
- Use preset and clear safely.
- Read a characteristic table.

M4.02 — Registers and shift registers (3 hours)

Concept and explanation

A register stores a group of bits. Shift registers move data left or right on clock pulses and may support serial-in/serial-out, serial-in/parallel-out, parallel-in/serial-out and parallel-in/parallel-out operation.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- Identify data direction.
- Explain serial and parallel transfer.
- Relate a register to temporary storage.

> **Engineering / workplace application:** Serial communication and delay lines use shift registers.

M4.03 — Counters and frequency division (3 hours)

Concept and explanation

Counters pass through a sequence of states on clock pulses. Ripple counters are simple but accumulate propagation delay; synchronous counters change stages together.

> **Malayalam support:** □ □□□□ □□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□ □□□□□□□□.

Study points

- Distinguish asynchronous and synchronous counters.
- Determine modulus from the state sequence.
- Explain reset and unused states.

M4.04 — Timing and sequential-system design (3 hours)

Concept and explanation

A sequential system is specified by states, inputs, outputs and transitions. Clock period, propagation delay, setup time and hold time determine reliable operation.

Malayalam support: □ □□□□ □□□□□□□□□□□□□□□□□□ English technical terms □□□□□□□□□ □□□□□□□□.

Study points

- Draw a state diagram.
- Check timing constraints.
- Use a reset state for predictable startup.

Module summary: Sequential design requires attention to timing, state transitions, setup, hold and reset behaviour.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

iew its labelled learning-map diagram.

Module 2: Boolean algebra

e module to view its labelled learning-map diagram.

Module 3: Combinational

e module to view its labelled learning-map diagram.

Module 4: Sequential

Open the module to view its labelled learning-map diagram.

Official Activities and Practical Requirements

Official activities / self-learning

- No separate activity list is provided in the supplied PDF.

Practical requirements / equipment

- No separate practical equipment list is provided in the supplied PDF.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- Series Test: 2 periods/assessment component as stated in the supplied syllabus; the supplied header does not state a separate CIE/ESE mark distribution.

Module-wise Questions and Answers

module-1 — Number systems and digital codes

1. Explain Number systems and base conversion. **[3 marks]**

A number system is defined by its radix and set of digits. Decimal, binary, octal and hexadecimal representations describe the same quantity with different symbols; repeated division or positional expansion provides systematic conversion.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Complements and binary arithmetic. **[3 marks]**

One’s complement and two’s complement provide signed representation and subtraction methods. Binary addition, subtraction, multiplication and division follow positional rules, with carry and borrow handled at each bit position.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Binary codes and data representation. **[3 marks]**

BCD represents decimal digits separately, Gray code changes one bit between adjacent values, and ASCII represents characters. Code selection depends on the required reliability, display, communication or storage function.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Boolean algebra and Karnaugh maps

4. Explain Boolean laws and logic gates. **[3 marks]**

Boolean variables and operators AND, OR and NOT form the algebra of switching circuits. De Morgan’s theorems are especially useful for changing a circuit between active-high and active-low forms.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

5. Explain Canonical forms, minterms and maxterms. **[3 marks]**

A truth table can be expressed as a sum of minterms or product of maxterms. Canonical forms provide a standard starting point for minimization and circuit implementation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Karnaugh-map simplification. **[3 marks]**

A Karnaugh map arranges adjacent cells in Gray-code order so groups of 1s or 0s reveal simplified terms. Groups must contain powers of two and may wrap around the map edges.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Combinational circuits

7. Explain Half and full adders. **[3 marks]**

A half adder adds two bits and produces sum and carry. A full adder adds two bits and an input carry, allowing multi-bit addition by cascading stages.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Subtractors and arithmetic circuits. **[3 marks]**

Half and full subtractors produce difference and borrow. Complement-based subtraction can also use adder hardware, which is why arithmetic circuits often combine addition and subtraction control.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

9. Explain Encoders, decoders and comparators. **[3 marks]**

An encoder converts one active input into a coded output, while a decoder activates one output for a coded input. A magnitude comparator indicates A greater than, equal to, or less than B.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Multiplexers and demultiplexers. **[3 marks]**

A multiplexer selects one of many data inputs using select lines. A demultiplexer routes one input to a selected output, making both useful for data routing and function generation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Sequential circuits

11. Explain Latches and flip-flops. **[3 marks]**

A latch is level-sensitive whereas a flip-flop is normally edge-triggered. SR, JK, D and T forms provide different ways to store and change a single bit.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Registers and shift registers. **[3 marks]**

A register stores a group of bits. Shift registers move data left or right on clock pulses and may support serial-in/serial-out, serial-in/parallel-out, parallel-in/serial-out and parallel-in/parallel-out operation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

13. Explain Counters and frequency division. **[3 marks]**

Counters pass through a sequence of states on clock pulses. Ripple counters are simple but accumulate propagation delay; synchronous counters change stages together.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Timing and sequential-system design. **[3 marks]**

A sequential system is specified by states, inputs, outputs and transitions. Clock period, propagation delay, setup time and hold time determine reliable operation.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Number systems and base conversion* with one relevant application. (5 marks)
2. **2.** Define or explain *Complements and binary arithmetic* with one relevant application. (5 marks)
3. **3.** Define or explain *Boolean laws and logic gates* with one relevant application. (5 marks)
4. **4.** Define or explain *Canonical forms, minterms and maxterms* with one relevant application. (5 marks)
5. **5.** Define or explain *Half and full adders* with one relevant application. (5 marks)
6. **6.** Define or explain *Subtractors and arithmetic circuits* with one relevant application. (5 marks)
7. **7.** Define or explain *Latches and flip-flops* with one relevant application. (5 marks)
8. **8.** Define or explain *Registers and shift registers* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Number systems and digital codes:** A reliable conversion method prevents errors when a digital quantity is represented in another base.
- **Boolean algebra and Karnaugh maps:** Simplification reduces gate count, delay and wiring in a practical digital circuit.

- **Combinational circuits:** A clean design proceeds from specification to truth table, simplification, circuit drawing and verification.
- **Sequential circuits:** Sequential design requires attention to timing, state transitions, setup, hold and reset behaviour.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Number systems and base conversion	A number system is defined by its radix and set of digits. Decimal, binary, octal and hexadecimal representations describe the same quantity with different symbols; repeated division or positional expansion provides systematic conversion.
Complements and binary arithmetic	One's complement and two's complement provide signed representation and subtraction methods. Binary addition, subtraction, multiplication and division follow positional rules, with carry and borrow handled at each bit position.
Binary codes and data representation	BCD represents decimal digits separately, Gray code changes one bit between adjacent values, and ASCII represents characters. Code selection depends on the required reliability, display, communication or storage function.
Boolean laws and logic gates	Boolean variables and operators AND, OR and NOT form the algebra of switching circuits. De Morgan's theorems are especially useful for changing a circuit between active-high and active-low forms.
Canonical forms, minterms and maxterms	A truth table can be expressed as a sum of minterms or product of maxterms. Canonical forms provide a standard starting point for minimization and circuit implementation.

Term	Short meaning
Karnaugh-map simplification	A Karnaugh map arranges adjacent cells in Gray-code order so groups of 1s or 0s reveal simplified terms. Groups must contain powers of two and may wrap around the map edges.
Half and full adders	A half adder adds two bits and produces sum and carry. A full adder adds two bits and an input carry, allowing multi-bit addition by cascading stages.
Subtractors and arithmetic circuits	Half and full subtractors produce difference and borrow. Complement-based subtraction can also use adder hardware, which is why arithmetic circuits often combine addition and subtraction control.
Encoders, decoders and comparators	An encoder converts one active input into a coded output, while a decoder activates one output for a coded input. A magnitude comparator indicates A greater than, equal to, or less than B.
Multiplexers and demultiplexers	A multiplexer selects one of many data inputs using select lines. A demultiplexer routes one input to a selected output, making both useful for data routing and function generation.
Latches and flip-flops	A latch is level-sensitive whereas a flip-flop is normally edge-triggered. SR, JK, D and T forms provide different ways to store and change a single bit.
Registers and shift registers	A register stores a group of bits. Shift registers move data left or right on clock pulses and may support serial-in/serial-out, serial-in/parallel-out, parallel-in/serial-out and parallel-in/parallel-out operation.
Counters and frequency division	Counters pass through a sequence of states on clock pulses. Ripple counters are simple but accumulate propagation delay; synchronous counters change stages together.
Timing and sequential-system design	A sequential system is specified by states, inputs, outputs and transitions. Clock period, propagation delay, setup time and hold time determine reliable operation.

Official References and Resources

References listed in the supplied syllabus

1. Digital Design, M. Morris Mano.
2. Fundamentals of Digital Circuits, A. Anand Kumar.

Official learning resources listed

- https://www.w3schools.com/

In-page Search

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